Job Prospects in for Data Science Students

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Job Prospects in Technology & AI

As a soon to be graduate with a Master's of Science in Data Science, I am interested in what job prospects look like, especially in this new age of Artificial Intelligence (AI). I am interested in determining if there are specific specializations within Data Science or AI centric fields that are the most in demand, or industries that are leading the pack in hiring, or the typical educational requirements required to land a job in Data Science or AI.

Business Problem

With the explosion of AI capabilities, and businesses wanting to take advantage of it, more than 57 major companies have explicitly laid off more than 377,000 workers since 2023 (DisplaceIndex, 2026). In 2026 alone, nearly 93,000 tech employees have been laid off due to AI with 111 Tech layoff events attributed to AI (Layoffs.fyi, 2026). For universities and for students who are in college or looking to go back for another degree they need to know where to focus for the best career prospects. Universities need to ensure they are offering the most relevant and up to date classes to provide a cutting edge education to attract students to their school. For students, whether they are looking for certificates, Bachelors, or Masters degrees, they want to know that they will pick a field that is still relevant and that they will have solid career prospects upon graduation. This study looks to explore these questions. What skills have the biggest payoff in the technology field? What job, by primary job title or category is seen most in reports for recruitment and hiring? Which jobs look to be the most in demand, by having the highest salaries and demand signals? Is there a specific AI or technical specialization that makes a prospective employee more valuable? Are there specific job roles that have higher career growth or job security?

Data Selection

I will be utilizing the AI Jobs Market 2025-2026 | Salaries Dataset that is available on Kaggle. This dataset was compiled from publicly available salary reports and job market research for 2025-2026. It includes data from Radio Global Salary Benchmark 2026, Robert Half Technology Salary Guide 2026, Motion Recruitment AI/ML Salary Report 2025, LinkedIn Workforce Insights 2025-2026, and Glassdoor AI Role Salary 2025. The salary figures are all in USD amounts and include base salary and standard bonuses. The emerging roles, such as, LLM Engineer, Rag Engineer, and AI Agent Developer were sourced from live job board aggregations on LinkedIn, Indeed, and Wellfound from 2025-2026. This dataset was last updated 7 months ago, so the data is still relatively new and should present good insights for hiring in the technology field. This dataset includes 1,500 job postings that cover 25 distinct Artificial Intelligence and Machine Learning roles that span 12 unique categories. The data comes from 13 countries and one Remote/Global category that spans 12 different industries and includes 5 company sizes from startups to big tech companies.

Exploratory Data Analysis

The analysis process will begin with validating the dataset for completeness. First, the shape of the data was validated, showing the total count at 1,500 rows and 25 columns. Next, column types were validated to ensure that any numerical values were represented as such instead of a string. Any values that need to be numeric and were actually strings would throw off the models. There is a mix of 9 integer columns, 4 float columns, and 12 categorial features. After checking types it was important to see if there were any missing values. If there were missing values I would need to decide how to handle them with either imputing or dropping. This check revealed

there were 0 missing values across any row or column in the dataset. This is a clean dataset and it is ready for modeling. I was interested in seeing the distribution of the education requirements and what the most sought after education was. This revealed a fairly uniform distribution for the different educational levels:

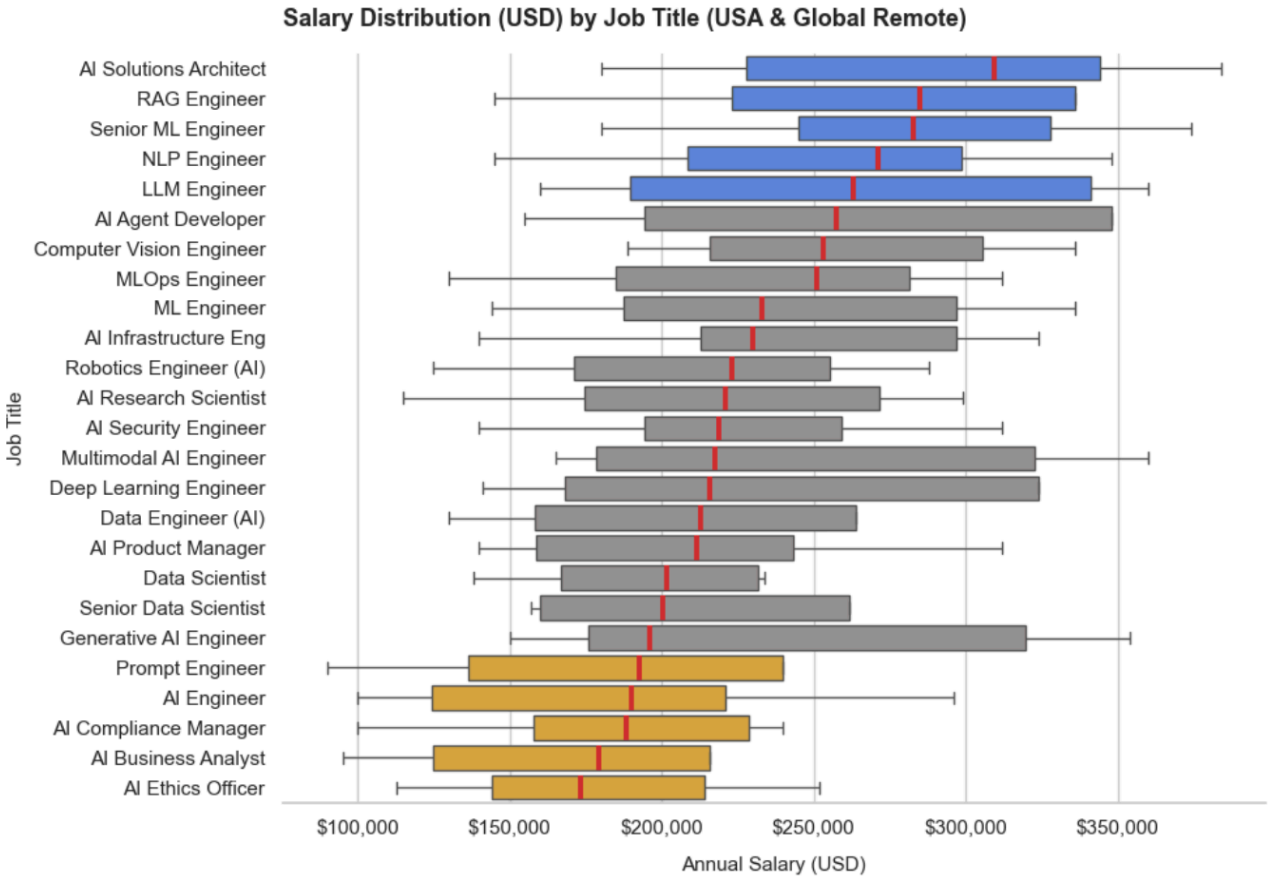
- Associate's: 20.97%
- Bootcamp/Self-taught: 20.39%
- Bachelor: 19.22%
- Master: 19.61%
- PhD: 19.81%

The Top Job roles, by count, were: AI Engineering, Data Science, Governance, Business, Data Engineering, Architecture, Robotics, Security, Infrastructure and ML Operations. The top hiring industries in order are: Automotive , Government, Finance, Consulting, Healthcare, Retail, Education, Energy, Research, and Media. As I am focused on available jobs for someone like me, graduating in the United States, I will narrow down the focus from this broad view of all the data, including all the countries to filtering the dataset to only assess job prospects for people in the United States or Global/Remote work. For USA and Global Remote there are 515 total records that will be utilized for this study.

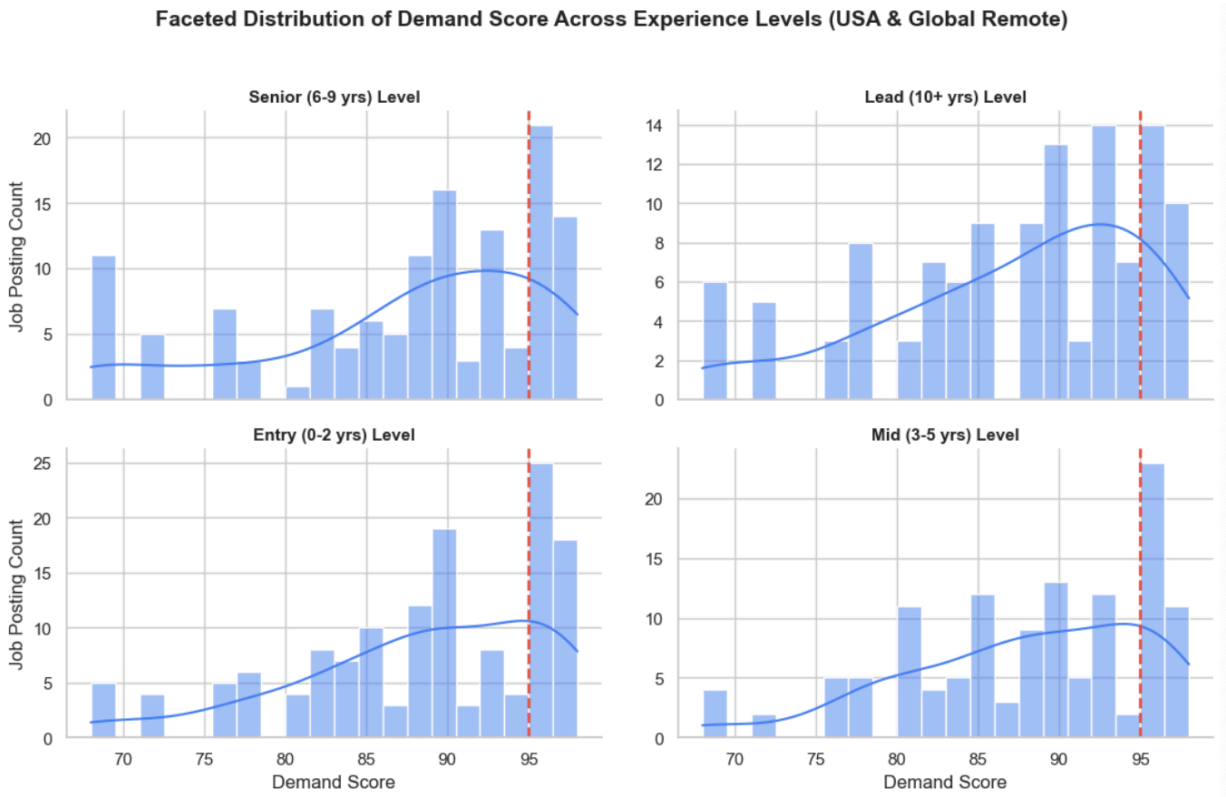
Visualizations

Visualizing the data helps in understanding additional details about the dataset so I started with viewing the salary distribution by job role. I did this with a box and whisker plot to view the distribution of salary by the different job roles. This would make it easy to see which jobs

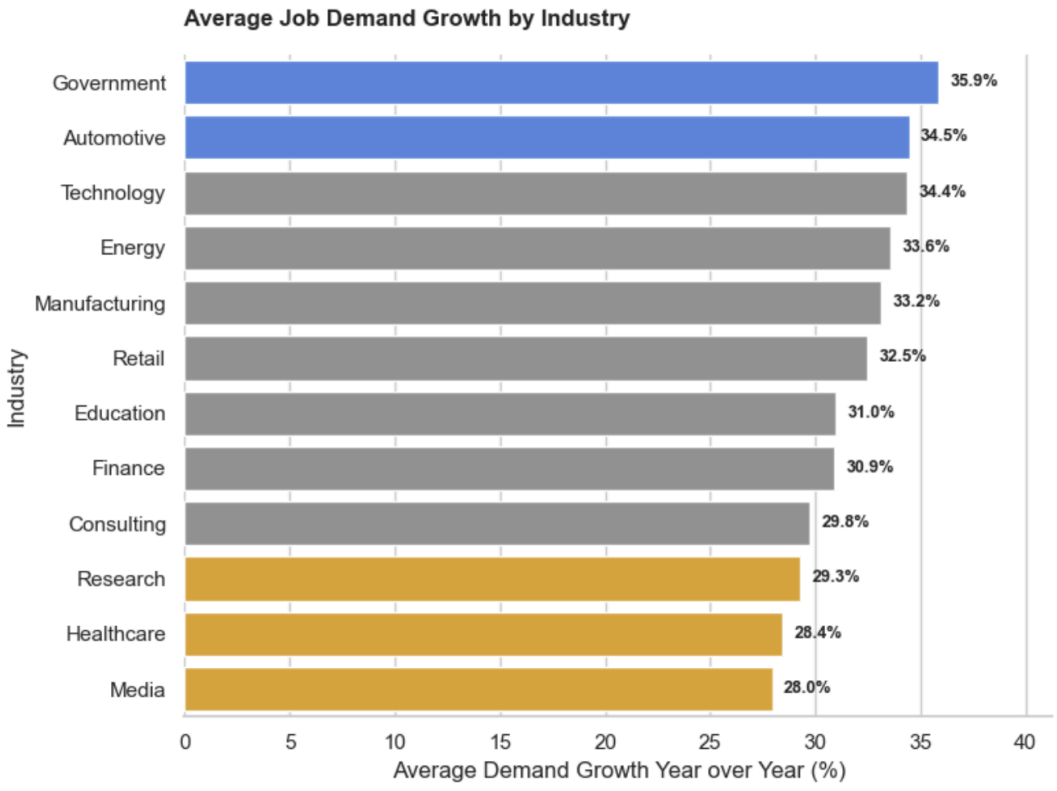
commanded the highest median salaries, which the lowest, and the general range for the salary band for each position type.



To see the shape and tendency of the data and to see if roles were concentrated around specific skill demands and to see the distribution and spread of the *demand_score* compared to experience levels and job roles I created histograms split by experience level.

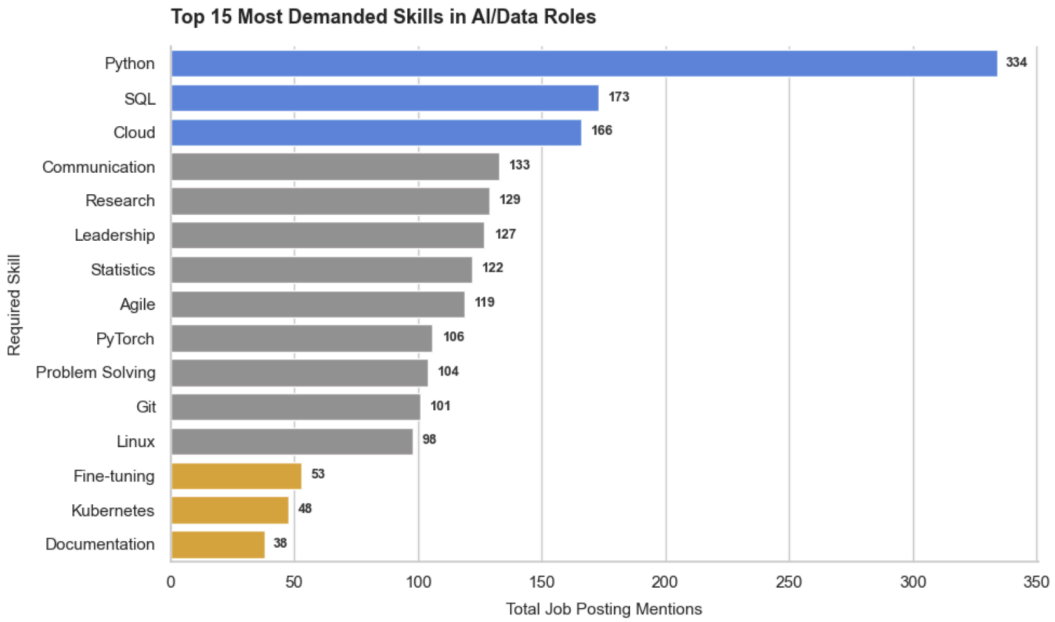


Next, I plotted total job openings by industry to be able to have a quick view of job demand growth across the different industries for comparison. I was interested which sectors were seeing the most growth for AI and tech jobs and which the least. This would help soon to be graduates and other job

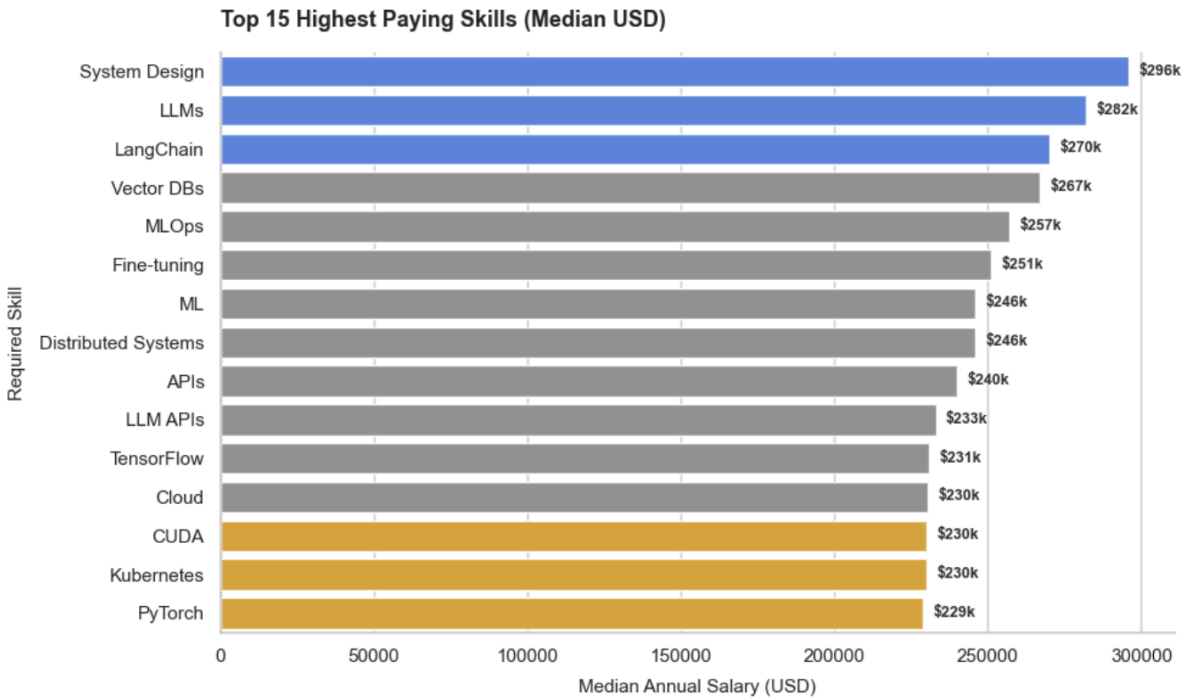


hunters to understand the best job sectors to chase and where to focus their job hunt.

Having the right skills to land a position is key, therefore I was interested in which skills were most frequently listed within a job posting. The number one most mentioned skill is Python, that appeared in 334 of the 515 USA and Global Remote job postings. This was followed by SQL at 173 postings, and Cloud at 166.



The highest paying skills, however, told a different story. Here, understanding System Design commanded the highest salary, for US and Global/Remote jobs at a staggering $296,000 as the annual median salary. This was followed closely by LLMs at $282,000 and LangChain at $270,000. On the top 15 list



PyTorch was the lowest at $229,000.

Modeling

There were a number of questions I was interested in being able to answer with this filtered USA and Global/Remote jobs dataset, which also meant that there was a wide range of modeling that could be done. For a fuller view of available jobs across the tech industry I will perform modeling to evaluate all positions and then narrow my focus to only data science.

Random Forest Regressor - Target Variable: *annual_salary_usd*

R-Squared: 0.6059 | RMSE: $46,103.46 | MAE: $38,265.60 | MAPE: 18.42%

This first model focused on being able to predict the annual salary for any job posting within the filtered (N=515) USA and Global Remote job postings. The R-Squared of 0.6059 means the model accounted for 60.59% of the overall variance in the base annual salary. This R-Squared score is just 10% better than a coin flip and demonstrates the predictive capability of the model. On average, the model's predictions were off from the actual salaries by $38,265.60. Validation of the Mean Absolute Percentage Error (MAPE), shows that there was an average error rate of 18.42% across the varying pay bands. The RMSE of $46,103.46 indicates there were some occasional outliers in salary. The top features that provided the model the most information on predicting salary included *is_senior* at 31.78%, and *years_of_experience* at 10.11%. This shows that time in service and senior level roles carried the most influence for the model. This was followed by technical skills such as, system design, at 6.73%, fine-tuning at 4.68%, documentation at 3.36% and leadership at 3.03% as the additional driving features for predicting salary.

Random Forest Regressor - Target Variable: *ai_salary_premium_pct*

R-Squared: 0.4016 | RMSE: 3.07% | MAE: 2.47%| MAPE: 23.58%

With the base salary predictions it was now possible to try to predict the percentage of the salary increase that different technical skills and experience levels that an AI role would command relative to the standard baseline roles. The R-Squared of 0.4016 shows that the model was able to explain 40.16% of the variance in the salary. The models predictions missed the true percentage premiums by an average of 2.47 percentage points, which indicates that the model was performing well. The MAPE of 23.58% shows the average prediction error across all the test samples was 23.58% The small RMSE of 3.07% show that the percentage increases commanded by AI positions were relatively tightly clustered with no large outliers that skewed the distribution. Here the biggest drivers for prediction was *years_of_experience* at 13.22% and then specialized AI/ML skills like ML algorithms at 8.50%, LLM APIs at 7.25% and hardware acceleration (CUDA) at 3.86% as the top skills that commanded the increased AI Salary Premium percentages.

K-Means Clustering

Silhouette coefficient: 0.2233 | Davies-Bouldin Index: 1.5809

To asses career growth, security and automation risks I utilized the K-Means Clustering focusing on cluster features of *demand_growth_yoy_pct, demand_score, ai_salary_premium_pct, annual_salary_usd,* and *is_llm_role.* The silhouette coefficient of 0.2233 indicates that there are moderately overlapping cluster boundaries which is typical for this type of labor market compensation rates and skill data. The Davies-Bouldin Index indicated that there were three profiles that emerged. Cluster 0 indicated a moderate YoY growth of 17.8% and a demand score of 80.4, while also commanding the highest AI salary premium of 13.9% and a strong salary of

$215,199. There were no LLM roles here, but instead the focus was on governance, compliance, business systems, and specialized hardware. The top roles included AI Business Analyst, Robotics Engineer, and AI compliance manager. Cluster 1 had a larger YoY growth of 21.4% as well as a high demand score of 88.9 with a base salary of $213,377 with the lowest AI salary premium of 7.8%. These positions included jobs such as Prompt Engineer, Deep Learning Engineer, and Senior Data Scientist. Cluster 2 had the highest YoY growth of 60.0%, the highest demand score at 95.2 and the highest average base salary of $253,190. The roles here were highly specialized in AI with almost half of them dedicated to LLM positions, as seen by the *llm_ratio* of 48.4%. The top roles included LLM Engineer, AI Agent Developer, and ML Engineer.

Regular Ridge Regression

R-Squared: 0.6271 | MAE: $18,736.80 | RMSE $22,428.59 | Optimal Alpha: 1.00 | Duan's Smearing Factor: 1.002113| Sample Size: 46

For the final model I wanted to narrow down the quest even further, to only include Data Science jobs, available within the filtered USA and Global Remote dataset, and to try to predict the salary. As the sample size was so small I utilized a Repeated K-Fold on the entire 46-row dataset to ensure that I would have enough predictive power for the model. I included over 50 different data shuffles and took the means for the Optimal alpha, Duan's Smearing Factor, MAE, RMSE and R-Squared. Salary was log-transformed to stabilize variance and Mincerian concavity was added to capture the diminishing marginal returns experienced in salary over a career. Back-transforming using Duan's Smearing Factor helps to correct bias and is widely used as the most accurate way to back transform log values. The averaged R-Squared of 0.6271 indicates that the

model was able to explain nearly 63% of the total variance in salary for a data scientist based only on experience, company size, and remote status. On average, the out of sample prediction missed the actual salary amounts by $18,736.80 (averaged), and the RMSE of $22,428.59 (averaged) confirms there were few outliers. The primary drivers for the model's predictions came from experience being a lead (0.192357), or having worked at a bigger company (0.132189), while small startups penalized salary (-0.119162), along with bootcamp or self-taught also penalizing salary (-0.054116) showing data scientists will make less money at a startup and if they don't have a formal degree.

Ethical Considerations

This dataset was compiled to provide an overview of AI related jobs that are available across different countries, industries and companies. As the focus of the study was to understand the job market available for US graduates the dataset was filtered to include only US and Global/ Remote jobs. That took the dataset down to only 515 individual datapoints. This data was compiled from publicly available salary and job market research and includes only five sources. There are a number of job posting boards, so this does not present an exhaustive list of all available AI jobs for the US and Global/ Remote. To have complete understanding then all available job posting sites should be catalogued and determine which sites are most widely used for technical job postings. Then, all of this data would need to be scraped and correlated to create a single datasource. This would be an exhaustive process to ensure a 100% possible view of all tech jobs was achieved. LinkedIn is widely used for job postings, so, for this study, I believe this is a fairly representative dataset.

This dataset was also last updated 7 months ago which means it may not precisely reflect the current state of technology and AI jobs as the landscape may have shifted. This is not too long of a time so I do believe this should still be a fairly close representation to current job postings.To mitigate these ethical concerns there must be a clear narrative to denote these limitations in the data and the modeling that was utilized to complete this dataset. Comparing the results from this study to current posted information for jobs in the tech industry is a good final step to validate the results. It is also unknown what type of bias or privacy concerns exist in this dataset. Students should not heavily rely on results from these predictions and students should not read the salaries as what they can expect to find in their local areas. Salaries rely heavily on the location in which the work is done or where the person resides and can vary widely from state to state. The near-uniform education and experience distributions suggest that these may have been synthetically created which most certainly applies additional concerns to the validity and real-world correctness of this data.

Final Summary

According to Futurewarns Research team, the fastest growing AI careers in 2026 (top 5 only) are AI/Machine Learning Engineer, Data Scientist, AI Product Manager, Prompt Engineer / AI Interaction Designer, and AI-Focused Information Security Analyst. The top 5 most in demand jobs for the USA and Global Remote filtered dataset included LLM Engineer, Senior ML Engineer, AI Agent Developer, RAG Engineer and ML Engineer. Looking at the results of the K-Means Clustering included the AI Compliance Manager (security), Prompt Engineer, LLM Engineer, and ML Engineer positions were all seeing growth. These look to be very closely aligned with what Futurewarns was predicting for the fastest growing AI careers for 2026.

For salary, Futurewarns lists as the typical US salary range as being comparable to what was seen in this dataset, with AI/ML engineer making 110-250K+ and a Data Scientist from 100-190K (2026). For USA and Global Remote positions the salary is a bit higher; for an ML Engineer it was a around 175-290K, while the Data Scientist was around 160-235K. Bifei Wang, on Extern, lists the fastest growing roles by percentage of growth to be AI and Machine Learning specialists, Data Analysts and Scientist, Software and application developers, both DevOps and Cloud engineers, and then Cybersecurity and Information Security Analysts (2025). The 15 highest paying skills, which equates to demand, included skills that would be used by each of these job positions, except the Cybersecurity.

I believe these models need additional tuning to be ready for deployment; they could help show which skills would be the most beneficial for students to learn and universities to focus on to ensure they are creating employable workers. Referring to the technical features/skills that drove salary for the Random Forest Regressor, as well as the additional AI skills that drove salary premiums would be advantageous for job seekers to know and understand. That would allow them to focus their skills on these top required skills, the job types that were seeing the most year over year growth, and which skills would gross them the highest salaries. A number of these skills are ones that Data Scientists should know, such as Python, machine learning algorithms, LLMs, fine-tuning skills, and statistics. It would be beneficial for Data Science students to take the AI/Generative AI classes to provide them with even more employable skills. It is also very clear that Python is one of the most demanded skills for AI/Data roles so universities should focus even more on ensuring they are pushing their students to be strong Python programmers.

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Appendix